

Street Vendor Discovery System

Minimal Functional Dependency Sets

DBMS Project | DA-IICT

A **Minimal FD Set (Canonical Cover)** is a simplified, equivalent set of functional dependencies that satisfies three conditions: **(1)** no redundant FDs, **(2)** no extraneous attributes on the left-hand side, and **(3)** a single attribute on every right-hand side. FDs marked **✓ Yes** belong to the minimal cover; FDs marked **—** are excluded because they represent alternate keys or dependencies derivable from the primary key.

1. City

#	Functional Dependency	In Minimal Cover?
1	City_ID → City_Name	✓ Yes
2	City_ID → State	✓ Yes
3	City_Name, State → City_ID	—

Minimal Cover: City_ID → City_Name | City_ID → State

2. Vendor

#	Functional Dependency	In Minimal Cover?
1	Vendor_ID → Vendor_Name	✓ Yes
2	Vendor_ID → Contact_No	✓ Yes
3	Vendor_ID → Years_of_Experience	✓ Yes
4	Vendor_ID → Average_Rating	✓ Yes
5	Vendor_ID → Registration_Date	✓ Yes
6	Vendor_ID → Role	✓ Yes
7	Contact_No → Vendor_ID	—

Minimal Cover: Vendor_ID → Vendor_Name | Vendor_ID → Contact_No | Vendor_ID → Years_of_Experience | Vendor_ID → Average_Rating | Vendor_ID → Registration_Date | Vendor_ID → Role

3. Stall

#	Functional Dependency	In Minimal Cover?
1	Stall_ID → Stall_Name	✓ Yes
2	Stall_ID → Location	✓ Yes
3	Stall_ID → Opening_Time	✓ Yes
4	Stall_ID → Closing_Time	✓ Yes
5	Stall_ID → Payment_Method	✓ Yes
6	Stall_ID → Is_Mobile	✓ Yes
7	Stall_ID → Seating_Available	✓ Yes
8	Stall_ID → Vendor_ID	✓ Yes
9	Stall_ID → City_ID	✓ Yes

Minimal Cover: Stall_ID → Stall_Name | Stall_ID → Location | Stall_ID → Opening_Time | Stall_ID → Closing_Time | Stall_ID → Payment_Method | Stall_ID → Is_Mobile | Stall_ID → Seating_Available | Stall_ID → Vendor_ID | Stall_ID → City_ID

4. Food_Category

#	Functional Dependency	In Minimal Cover?
1	Category_ID → Category_Name	✓ Yes
2	Category_ID → Description	✓ Yes
3	Category_Name → Category_ID	—

Minimal Cover: Category_ID → Category_Name | Category_ID → Description

5. Food_Item

#	Functional Dependency	In Minimal Cover?
1	Food_ID → Food_Name	✓ Yes
2	Food_ID → Description	✓ Yes
3	Food_ID → Category_ID	✓ Yes
4	Food_Name → Food_ID	—

Minimal Cover: Food_ID → Food_Name | Food_ID → Description | Food_ID → Category_ID

6. Menu

#	Functional Dependency	In Minimal Cover?
1	Menu_ID → Stall_ID	✓ Yes
2	Stall_ID → Menu_ID	✓ Yes

Minimal Cover: Menu_ID → Stall_ID | Stall_ID → Menu_ID

7. Menu_Item

#	Functional Dependency	In Minimal Cover?
1	(Menu_ID, Food_ID) → Price	✓ Yes
2	(Menu_ID, Food_ID) → Availability	✓ Yes
3	(Menu_ID, Food_ID) → Dietary_Tag	✓ Yes
4	(Menu_ID, Food_ID) → Is_Seasonal	✓ Yes

Minimal Cover: (Menu_ID, Food_ID) → Price | (Menu_ID, Food_ID) → Availability | (Menu_ID, Food_ID) → Dietary_Tag | (Menu_ID, Food_ID) → Is_Seasonal

8. Customer

#	Functional Dependency	In Minimal Cover?
1	Customer_ID → First_Name	✓ Yes
2	Customer_ID → Last_Name	✓ Yes
3	Customer_ID → Email	✓ Yes
4	Customer_ID → Phone	✓ Yes
5	Customer_ID → Registration_Date	✓ Yes
6	Customer_ID → Role	✓ Yes
7	Email → Customer_ID	—

Minimal Cover: Customer_ID → First_Name | Customer_ID → Last_Name | Customer_ID → Email | Customer_ID → Phone | Customer_ID → Registration_Date | Customer_ID → Role

9. Review

#	Functional Dependency	In Minimal Cover?
1	Review_ID → Rating	✓ Yes
2	Review_ID → Comment	✓ Yes
3	Review_ID → Review_Date	✓ Yes
4	Review_ID → Visit_Count	✓ Yes
5	Review_ID → Customer_ID	✓ Yes
6	Review_ID → Stall_ID	✓ Yes

Minimal Cover: Review_ID → Rating | Review_ID → Comment | Review_ID → Review_Date | Review_ID → Visit_Count | Review_ID → Customer_ID | Review_ID → Stall_ID

10. Authority

#	Functional Dependency	In Minimal Cover?
1	Authority_ID → Name	✓ Yes
2	Authority_ID → Department	✓ Yes
3	Authority_ID → Contact	✓ Yes
4	Authority_ID → Role	✓ Yes

Minimal Cover: Authority_ID → Name | Authority_ID → Department | Authority_ID → Contact | Authority_ID → Role

11. License

#	Functional Dependency	In Minimal Cover?
1	License_ID → License_Type	✓ Yes
2	License_ID → Issue_Date	✓ Yes
3	License_ID → Expiry_Date	✓ Yes
4	License_ID → Status	✓ Yes
5	License_ID → Vendor_ID	✓ Yes
6	License_ID → Authority_ID	✓ Yes

Minimal Cover: License_ID → License_Type | License_ID → Issue_Date | License_ID → Expiry_Date | License_ID → Status | License_ID → Vendor_ID | License_ID → Authority_ID

12. Inspection

#	Functional Dependency	In Minimal Cover?
1	Inspection_ID → Inspection_Date	✓ Yes
2	Inspection_ID → Hygiene_Rating	✓ Yes
3	Inspection_ID → Remarks	✓ Yes
4	Inspection_ID → Violation_History	✓ Yes
5	Inspection_ID → Stall_ID	✓ Yes
6	Inspection_ID → Authority_ID	✓ Yes

Minimal Cover: Inspection_ID → Inspection_Date | Inspection_ID → Hygiene_Rating | Inspection_ID → Remarks | Inspection_ID → Violation_History | Inspection_ID → Stall_ID | Inspection_ID → Authority_ID